

May 2nd Week Progress Update – 2026

Dear Sir,

I hope you are doing well. I am writing to provide an update on my progress for May 2nd week 2026 and outline my planned objectives for May leading up to the midyear evaluation.

Work Completed in May 2nd Week (May 6 to 13):

- Completed verification of the Keccak implementation for both SHAKE-128 and SHAKE-256 (100% functional coverage), verifying all operational modes, message lengths, and XOF output lengths.
- Implemented a 2-stage pipelined architecture in the Keccak design and restructured the architecture to include 4 separate Keccak processing channels, each with its own dedicated core.
- Studied recent research papers that adopt similar multi-core high-throughput architectures.
- Focused the design modifications toward maximizing throughput and overall performance. Identified that the future Dilithium controller can utilize all 4 channels during different operations to justify the increased area usage while improving throughput.
- Completed synthesis of the design on Intel Cyclone V FPGA and achieved approximately 84 MHz Fmax for single-core operation (minor setup timing violations observed).
- Planned next step: begin implementation of the Number Theoretic Transform (NTT).
- Timing optimization and Fmax improvements for the Keccak design will be revisited later if necessary.

Goals for 3rd Week of May (May 14 to 20):

- Research and literature review on NTT for high-performance and efficient design.
- Hardware design and UVM verification.
- Start synthesis for NTT.
- Start thesis writing.

Goals Left Before Midyear Evaluation:

- Complete NTT implementation.
- Strengthen literature review by identifying new optimization approaches from recent research.
- Structured thesis writing.

Please let me know if you would like any adjustments to the plan or additional focus areas.

Best regards,
Muhammad Abdul Ahad